

MGT 403:
Probability and Statistics

Lecture 4
Binomial Distribution

Tossing a Coin One Time

- Assume you have a coin that comes up heads with probability p



$$\mathbf{P}(H) = p$$

$$\mathbf{P}(T) = 1 - p$$

Tossing a Coin n Times

- Assume you have n of these coins
- Let **X** = number of heads when you toss n coins



- What is the probability distribution of X ?

Tossing $n = 1$ Coin



$$\mathbf{P}(X = 0) = \mathbf{P}(T) = 1 - p$$

$$\mathbf{P}(X = 1) = \mathbf{P}(H) = p$$

Tossing $n = 2$ Coins



$$\mathbf{P}(X = 0) = \mathbf{P}(TT) = (1 - p)^2$$

$$\mathbf{P}(X = 1) = \mathbf{P}(HT) + \mathbf{P}(TH) = 2p(1 - p)$$

$$\mathbf{P}(X = 2) = \mathbf{P}(HH) = p^2$$

Tossing $n=3$ Coins



$$\mathbf{P}(X = 0) = \mathbf{P}(TTT) = (1 - p)^3$$

$$\mathbf{P}(X = 1) = \mathbf{P}(HTT) + \mathbf{P}(THT) + \mathbf{P}(TTH) = 3p(1 - p)^2$$

$$\mathbf{P}(X = 2) = \mathbf{P}(HHT) + \mathbf{P}(HTH) + \mathbf{P}(THH) = 3p^2(1 - p)$$

$$\mathbf{P}(X = 3) = \mathbf{P}(HHH) = p^3$$

Tossing n Coins

- **P(X=k) is found by adding up the probability of every outcome with k heads and n-k tails**

$$\mathbf{P}(X = k) = \frac{n!}{k!(n - k)!} p^k (1 - p)^{n - k}$$

Factorials !

- **k!** means “k factorial”

$$k! = 1 \cdot 2 \cdot 3 \dots \cdot k$$

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- **Examples**

$$1! = 1$$

$$2! = 1 \cdot 2 = 2$$

$$3! = 1 \cdot 2 \cdot 3 = 6$$

Factorials !

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- **Examples**

$$0! = 1$$

$$1! = 1$$

$$2! = 1 \cdot 2 = 2$$

$$3! = 1 \cdot 2 \cdot 3 = 6$$

Factorials !

- **Examples**

$$\left(\frac{1}{2}\right)! =$$

Factorials !

- **Examples**

$$\left(\frac{1}{2}\right)! = \frac{\sqrt{\pi}}{2}$$

Binomial Distribution

- X = number of successes (heads) in n trials (tosses) = **binomial random variable**
- $P(X)$ is the **binomial distribution**

$$P(X = k) = \frac{n!}{k!(n - k)!} p^k (1 - p)^{n - k}$$





Example: Tossing a Fair Coin

Problem Toss a fair coin 10 times

- a** What is $P(X = 0)$?
- b** What is $P(X = 10)$?
- c** What is $P(X = 5)$?



1 2 3 4 5 6 7 8 9 10

Properties of Binomial Distribution

- X is a **binomial random variable** with n trials and success probability p
- **Properties of X**

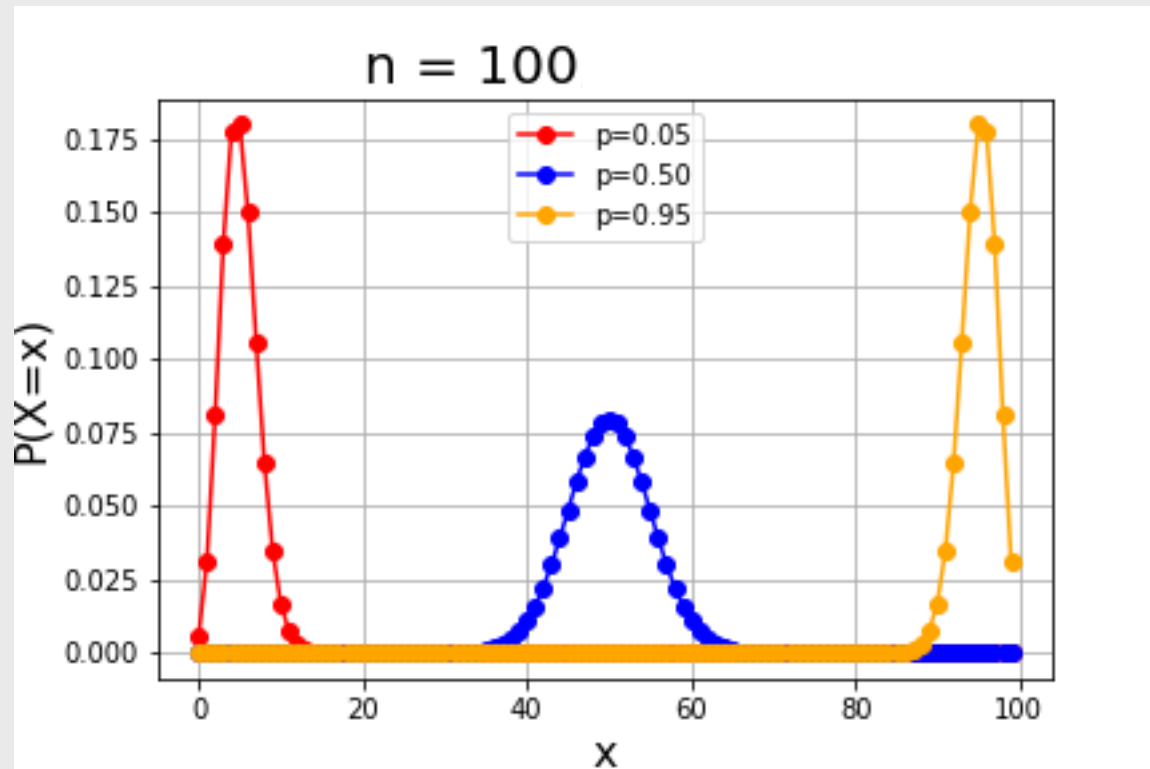
$$E(X) = np$$

$$\text{Var}(X) = np(1 - p)$$

$$\text{StDev}(X) = \sqrt{np(1 - p)}$$

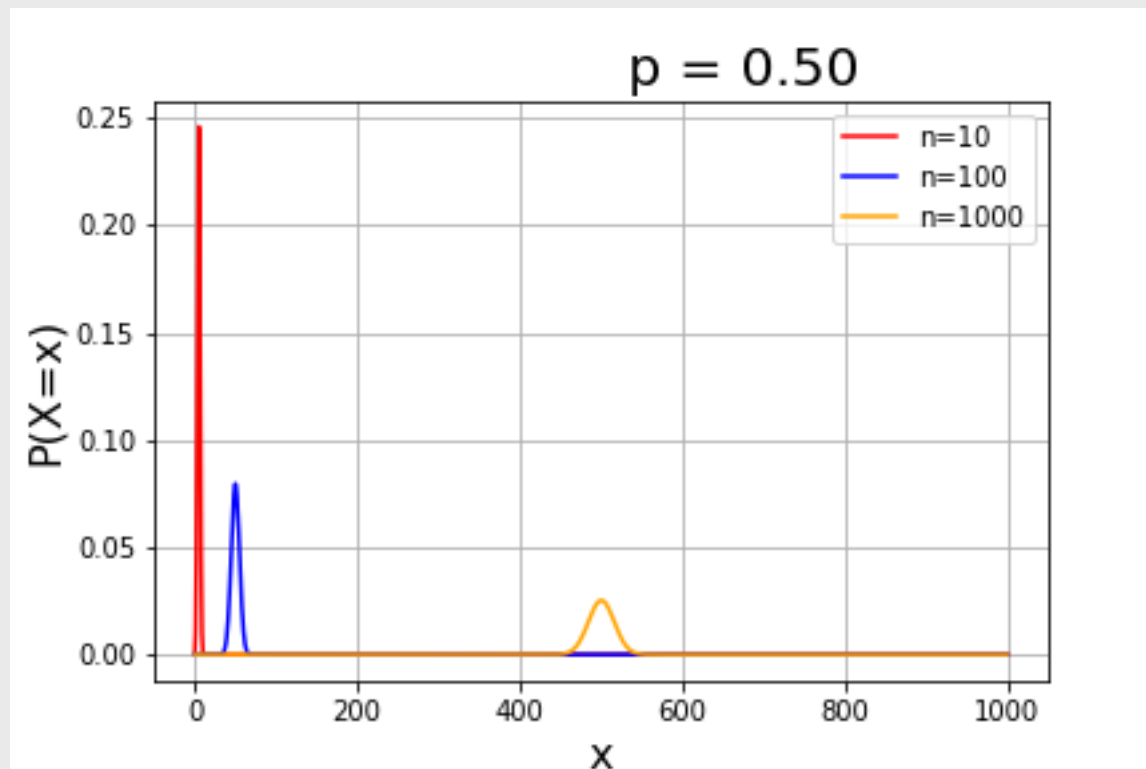
Visualizing the Binomial Distribution

- Plot of binomial distribution for different values of p



Visualizing the Binomial Distribution

- Plot of binomial distribution for different values of **n**



Binomial Distribution with Excel

- We can use Excel (and many other software) to calculate $P(X = k)$ and $P(X \leq k)$ for the binomial distribution

Binomial Distribution with Excel

- We can use Excel (and many other software) to calculate $P(X = k)$ and $P(X \leq k)$ for the binomial distribution

$$P(X = k) = \text{BINOM.DIST}(k, n, p, \text{FALSE})$$

$$P(X \leq k) = \text{BINOM.DIST}(k, n, p, \text{TRUE})$$



Binomial Distribution with Excel

- Different types of probabilities you can calculate

$$\begin{aligned}P(X > k) &= 1 - P(X \leq k) \\&= 1 - \text{BINOM.DIST}(k, n, p, \text{TRUE})\end{aligned}$$

$$\begin{aligned}P(X \geq k) &= 1 - P(X \leq k - 1) \\&= 1 - \text{BINOM.DIST}(k-1, n, p, \text{TRUE})\end{aligned}$$

$$\begin{aligned}P(l \leq X \leq k) &= P(X \leq k) - P(X \leq l - 1) \\&= \text{BINOM.DIST}(k, n, p, \text{TRUE}) - \\&\quad \text{BINOM.DIST}(l-1, n, p, \text{TRUE})\end{aligned}$$



Open heart surgery

- Problem** Nelly is an open heart surgeon at Yale New Haven Hospital, the probability that an individual patient does not survive a surgery is 0.05, and the outcome of each of her surgeries are independent. Nelly performs 20 surgeries over a 3-month period.
- a** What is the expected number of patients who do not survive?
 - b** What is the probability that 2 or more of her 20 patients do not survive?

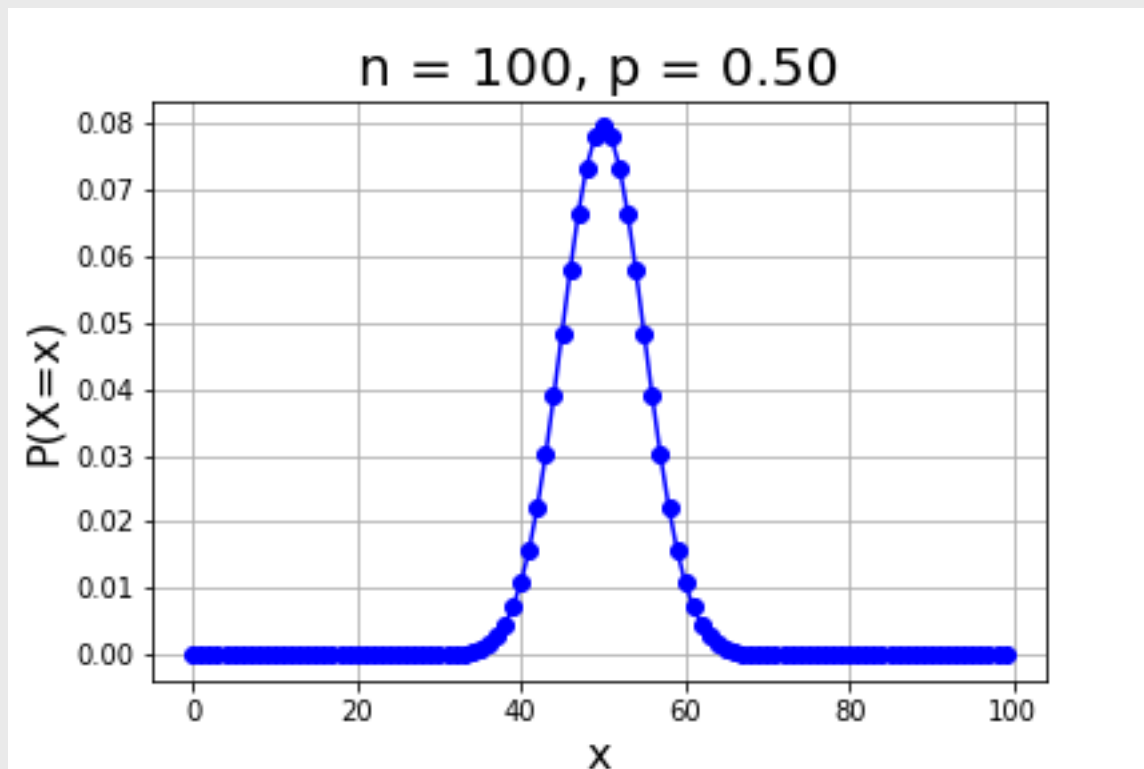


Schlitz live TV taste test

- Problem** Schlitz plans to assemble 100 people who all have stated that they are Budweiser drinkers and ask them to do a blind test comparison between Budweiser and Schlitz during a Superbowl break to be broadcasted on live TV. Based on tests, they have found that pretty much no-one are able to tell the difference between the two.
- a What is the expected number of the 100 persons who will choose Schlitz as their preference?
 - b What is the probability that at least 40 people will choose Schlitz as their preference?
 - c What is the probability that between 44 and 56 will choose Schlitz as their preference?

Schlitz Line TV Test

- X = number who like Schlitz





Going Pro

- **Z-Listo is an aspiring Clash Royale player. He needs a win rate greater than 0.9 to go pro. His actual win rate is $p=0.8$. Assume each game is independent**
 - a) What is the probability he goes pro if he is judged based on $n = 1$ games?**
 - b) What is the probability he goes pro if he is judged based on $n = 10$ games?**
 - c) What is the probability he goes pro if he is judged based on $n = 100$ games?**

Law of Large Numbers

- Perform n independent trials and the probability of success on each trial = p
- X = number of successes in n trials
- The Law of Large Numbers says

$$\text{As } n \rightarrow \infty, \quad \frac{X}{n} \rightarrow p$$

Law of Large Numbers

- Intuition: X/n is a random variable
- X/n has a mean and standard deviation

Law of Large Numbers

- The mean of X/n is

$$\begin{aligned}\mathbf{E} \left[\frac{X}{n} \right] &= \frac{\mathbf{E}[X]}{n} \\ &= \frac{np}{n} \\ &= p\end{aligned}$$

Law of Large Numbers

- The variance of X/n is

$$\begin{aligned}\text{Var}\left(\frac{X}{n}\right) &= \frac{\text{Var}(X)}{n^2} \\ &= \frac{np(1-p)}{n^2} \\ &= \frac{p(1-p)}{n}\end{aligned}$$

Law of Large Numbers

- The standard deviation of X/n is

$$\begin{aligned}\sigma &= \sqrt{\text{Var} \left(\frac{X}{n} \right)} \\ &= \sqrt{\frac{p(1-p)}{n}}\end{aligned}$$

Law of Large Numbers

- **The mean of $X/n = p$ (the true probability)**
- **The standard deviation of X/n gets smaller with more trials**
- **As you do more trials, the uncertainty in X/n goes to zero**



Independent Trials



- **Donald Trump faces seven scheduled criminal and civil trials. He is also running for president, which gives him a busy schedule. For each trial, he will show up with probability 0.5. If he does not show up, he will be convicted with probability 0.9. If he shows up, he will lower the chance of conviction to 0.8, either by charming or intimidating the jury. All trials are independent because justice is blind.**
 - a) **What is the probability that Donald Trump is acquitted in all trials?**
 - b) **What is the probability that Donald Trump is acquitted in all trials if he shows up to all trials?**

Next Time

- **Normal distribution** – this probability distribution describes almost everything we deal with that is “normal”
- Also known as the “bell curve”
- We get to use “normal tables”! ☹️